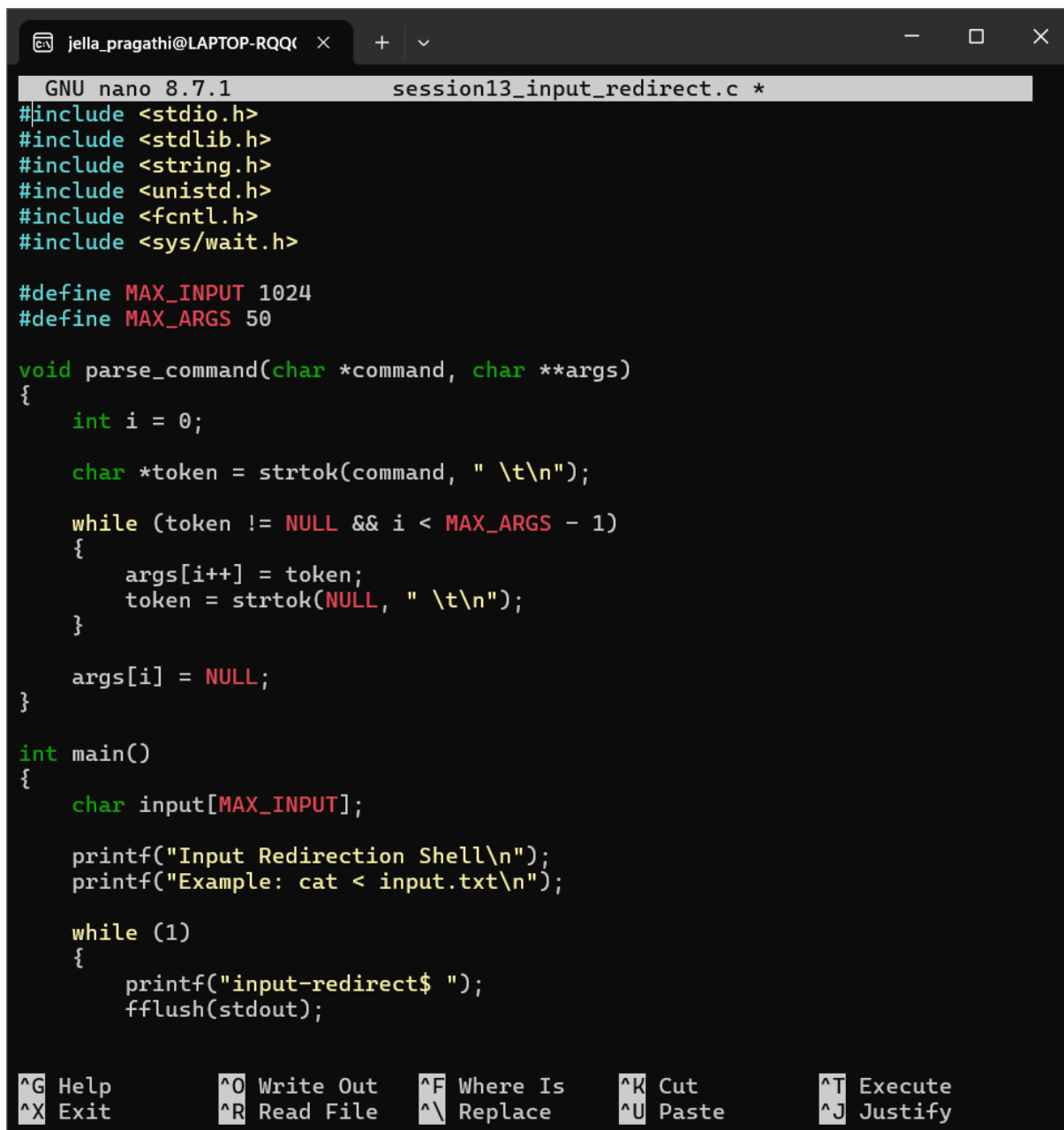


Skill Week 7

Session 13

Task 1

To Parse Input Redirection, Open Files, Redirect stdin, Handle Missing Files, Restore Streams, Test File Inputs.



```
GNU nano 8.7.1 session13_input_redirect.c *
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <fcntl.h>
#include <sys/wait.h>

#define MAX_INPUT 1024
#define MAX_ARGS 50

void parse_command(char *command, char **args)
{
    int i = 0;

    char *token = strtok(command, " \\t\\n");

    while (token != NULL && i < MAX_ARGS - 1)
    {
        args[i++] = token;
        token = strtok(NULL, " \\t\\n");
    }

    args[i] = NULL;
}

int main()
{
    char input[MAX_INPUT];

    printf("Input Redirection Shell\\n");
    printf("Example: cat < input.txt\\n");

    while (1)
    {
        printf("input-redirect$ ");
        fflush(stdout);
```

^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\\ Replace ^U Paste ^J Justify

```
    if (fgets(input, sizeof(input), stdin) == NULL)
        break;

    input[strcspn(input, "\n")] = '\0';

    if (strcmp(input, "exit") == 0)
        break;

    char *redirect = strchr(input, '<');

    if (redirect == NULL)
    {
        printf("Error: Use '<' for input redirection.\n");
        continue;
    }

    *redirect = '\0';

    char command[MAX_INPUT];
    char filename[MAX_INPUT];

    strcpy(command, input);
    strcpy(filename, redirect + 1);

    char *file = strtok(filename, " \t\n");

    if (file == NULL)
    {
        printf("Error: No input file specified.\n");
        continue;
    }

    char *args[MAX_ARGS];

    | parse_command(command, args);
```

^G Help
^X Exit

^O Write Out
^R Read File

^F Where Is
^_ Replace

^K Cut
^U Paste

^T Execute
^J Justify

```
jella_pragathi@LAPTOP-RQQ( x + v - □ ×
GNU nano 8.7.1 session13_input_redirect.c *
    if (args[0] == NULL)
    {
        printf("Error: No command specified.\n");
        continue;
    }

    pid_t pid = fork();

    if (pid < 0)
    {
        perror("fork");
        continue;
    }

    if (pid == 0)
    {
        int fd = open(file, O_RDONLY);

        if (fd < 0)
        {
            perror("open");
            exit(EXIT_FAILURE);
        }

        if (dup2(fd, STDIN_FILENO) < 0)
        {
            perror("dup2");
            close(fd);
            exit(EXIT_FAILURE);
        }

        close(fd);

        execvp(args[0], args);

        perror("execvp");
        exit(EXIT_FAILURE);
    }

    |
```

^G Help	^O Write Out	^F Where Is	^K Cut	^T Execute
^X Exit	^R Read File	^I Replace	^U Paste	^J Justify

```
        perror("execvp");
        exit(EXIT_FAILURE);
    }
    else
    {
        waitpid(pid, NULL, 0);
    }
}

printf("Input redirection shell terminated.\n");

return 0;
}
|
```

^G Help	^O Write Out	^F Where Is	^K Cut	^T Execute
^X Exit	^R Read File	^I Replace	^U Paste	^J Justify

```
jella_pragathi@LAPTOP-RQQT555:~/OSSP$ cd ~/OSSP
jella_pragathi@LAPTOP-RQQT555:~/OSSP$ nano session13_input_redirect.c
jella_pragathi@LAPTOP-RQQT555:~/OSSP$ gcc -Wall -Wextra -g session13_input_
redirect.c -o session13_input_redirect
jella_pragathi@LAPTOP-RQQT555:~/OSSP$ echo "Hello from input file!" > input
.txt
jella_pragathi@LAPTOP-RQQT555:~/OSSP$ cat input.txt
Hello from input file!
jella_pragathi@LAPTOP-RQQT555:~/OSSP$ ./session13_input_redirect
Input Redirection Shell
Example: cat < input.txt
input-redirect$ cat < input.txt
Hello from input file!
input-redirect$ wc -w < input.txt
4
input-redirect$ exit
Input redirection shell terminated.
jella_pragathi@LAPTOP-RQQT555:~/OSSP$ ./session13_input_redirect
Input Redirection Shell
Example: cat < input.txt
input-redirect$ cat < missing.txt
open: No such file or directory
input-redirect$ exit
Input redirection shell terminated.
jella_pragathi@LAPTOP-RQQT555:~/OSSP$ |
```

Task 2

To Parse Output Redirection, Create Output Files, Redirect stdout, Handle Permissions, Verify Output Data, Test Redirection.

```
jella_pragathi@LAPTOP-RQQ... session13_output_redirect.c *
GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <fcntl.h>
#include <sys/wait.h>

#define MAX_INPUT 1024
#define MAX_ARGS 50

void parse_command(char *command, char **args)
{
    int i = 0;

    char *token = strtok(command, " \\t\\n");

    while (token != NULL && i < MAX_ARGS - 1)
    {
        args[i++] = token;
        token = strtok(NULL, " \\t\\n");
    }

    args[i] = NULL;
}

int main()
{
    char input[MAX_INPUT];

    printf("Output Redirection Shell\\n");
    printf("Example: ls -l > output.txt\\n");

    while (1)
    {
        printf("output-redirect$ ");
        fflush(stdout);
    }
}
```

^G Help	^O Write Out	^F Where Is	^K Cut	^T Execute
^X Exit	^R Read File	^\\ Replace	^U Paste	^J Justify

```
if (fgets(input, sizeof(input), stdin) == NULL)
    break;

input[strcspn(input, "\n")] = '\0';

if (strcmp(input, "exit") == 0)
    break;

char *redirect = strchr(input, '>');

if (redirect == NULL)
{
    printf("Error: Use '>' for output redirection.\n");
    continue;
}

*redirect = '\0';

char command[MAX_INPUT];
char filename[MAX_INPUT];

strcpy(command, input);
strcpy(filename, redirect + 1);

char *file = strtok(filename, " \t\n");

if (file == NULL)
{
    printf("Error: No output file specified.\n");
    continue;
}

char *args[MAX_ARGS];

parse_command(command, args);
```

^G Help
^X Exit

^O Write Out
^R Read File

^F Where Is
^_ Replace

^K Cut
^U Paste

^T Execute
^J Justify

```
jella_pragathi@LAPTOP-RQQQ x + v
GNU nano 8.7.1 session13_output_redirect.c *

    if (args[0] == NULL)
    {
        printf("Error: No command specified.\n");
        continue;
    }

    pid_t pid = fork();

    if (pid < 0)
    {
        perror("fork");
        continue;
    }

    if (pid == 0)
    {
        int fd = open(file, O_WRONLY | O_CREAT | O_TRUNC, 0644);

        if (fd < 0)
        {
            perror("open");
            exit(EXIT_FAILURE);
        }

        if (dup2(fd, STDOUT_FILENO) < 0)
        {
            perror("dup2");
            close(fd);
            exit(EXIT_FAILURE);
        }

        close(fd);

        execvp(args[0], args);

        |
        perror("execvp");

    ^G Help    ^O Write Out  ^F Where Is   ^K Cut        ^T Execute
    ^X Exit    ^R Read File  ^\ Replace    ^U Paste      ^J Justify
```

```
        perror("execvp");
        exit(EXIT_FAILURE);
    }
    else
    {
        waitpid(pid, NULL, 0);
    }
}

printf("Output redirection shell terminated.\n");

return 0;
}
|

    ^G Help    ^O Write Out  ^F Where Is   ^K Cut        ^T Execute
    ^X Exit    ^R Read File  ^\ Replace    ^U Paste      ^J Justify
```

```
jella_pragathi@LAPTOP-RQQ... X + v
jella_pragathi@LAPTOP-RQQGT555:~/OSSP$ cat output.txt
total 452
-rw-r--r-- 1 jella_pragathi jella_pragathi 0 Aug 17 04:54 LICENSE
-rw-r--r-- 1 jella_pragathi jella_pragathi 265 Aug 17 05:03 Makefile
-rw-r--r-- 1 jella_pragathi jella_pragathi 0 Aug 17 04:54 README.md
drwxr-xr-x 2 jella_pragathi jella_pragathi 4096 Aug 1 08:02 assets
drwxr-xr-x 2 jella_pragathi jella_pragathi 4096 Aug 17 05:05 bin
drwxr-xr-x 2 jella_pragathi jella_pragathi 4096 Aug 1 08:01 docs
-rwxr-xr-x 1 jella_pragathi jella_pragathi 609 Aug 26 03:06 double_quotes.sh
drwxr-xr-x 2 jella_pragathi jella_pragathi 4096 Aug 1 08:01 include
-rw-r--r-- 1 jella_pragathi jella_pragathi 23 Sep 17 06:24 input.txt
drwxr-xr-x 2 jella_pragathi jella_pragathi 4096 Aug 17 05:05 obj
-rw-r--r-- 1 jella_pragathi jella_pragathi 0 Sep 17 06:29 output.txt
-rwxr-xr-x 1 jella_pragathi jella_pragathi 18248 Aug 31 03:47 pipeline
-rw-r--r-- 1 jella_pragathi jella_pragathi 1181 Aug 31 03:46 pipeline.c
-rwxr-xr-x 1 jella_pragathi jella_pragathi 19280 Aug 31 03:41 producer_consumer
-rw-r--r-- 1 jella_pragathi jella_pragathi 1865 Aug 31 03:41 producer_consumer.c
drwxr-xr-x 2 jella_pragathi jella_pragathi 4096 Aug 1 08:02 scripts
-rwxr-xr-x 1 jella_pragathi jella_pragathi 20752 Sep 16 08:19 session10_export
-rw-r--r-- 1 jella_pragathi jella_pragathi 1970 Sep 16 08:18 session10_export.c
-rwxr-xr-x 1 jella_pragathi jella_pragathi 19664 Sep 16 08:12 session10_pwd_exit
-rw-r--r-- 1 jella_pragathi jella_pragathi 963 Sep 16 08:11 session10_pwd_exit.c
-rwxr-xr-x 1 jella_pragathi jella_pragathi 20288 Sep 17 05:57 session11_history
-rw-r--r-- 1 jella_pragathi jella_pragathi 2099 Sep 17 05:57 session11_history.c
-rwxr-xr-x 1 jella_pragathi jella_pragathi 20840 Sep 17 06:04 session11_pipeline
-rw-r--r-- 1 jella_pragathi jella_pragathi 2792 Sep 17 06:04 session11_pipeline.c
-rwxr-xr-x 1 jella_pragathi jella_pragathi 20864 Sep 17 06:11 session12_multi_pipe
-rw-r--r-- 1 jella_pragathi jella_pragathi 3180 Sep 17 06:11 session12_multi_pipe.c
-rwxr-xr-x 1 jella_pragathi jella_pragathi 19800 Sep 17 06:08 session12_pipes
-rw-r--r-- 1 jella_pragathi jella_pragathi 1010 Sep 17 06:08 session12_pipes.c
-rwxr-xr-x 1 jella_pragathi jella_pragathi 20816 Sep 17 06:23 session13_input_redirect
-rw-r--r-- 1 jella_pragathi jella_pragathi 2295 Sep 17 06:23 session13_input_redirect.c
-rwxr-xr-x 1 jella_pragathi jella_pragathi 20816 Sep 17 06:28 session13_output_redirect
-rw-r--r-- 1 jella_pragathi jella_pragathi 2329 Sep 17 06:28 session13_output_redirect.c
-rwxr-xr-x 1 jella_pragathi jella_pragathi 16392 Aug 23 16:14 session4_wait
-rw-r--r-- 1 jella_pragathi jella_pragathi 1507 Aug 23 16:14 session4_wait.c
-rwxr-xr-x 1 jella_pragathi jella_pragathi 20344 Sep 16 07:00 session9_cd
-rw-r--r-- 1 jella_pragathi jella_pragathi 1913 Sep 16 07:00 session9_cd.c
-rwxr-xr-x 1 jella_pragathi jella_pragathi 21184 Sep 16 07:07 session9_dispatch
-rw-r--r-- 1 jella_pragathi jella_pragathi 2359 Sep 16 07:07 session9_dispatch.c
-rwxr-xr-x 1 jella_pragathi jella_pragathi 446 Aug 26 03:02 single_quotes.sh
```

```
-rw-r--r-- 1 jella_pragathi jella_pragathi 807 Aug 26 03:29 task2_encrypt.c
-rwxr-xr-x 1 jella_pragathi jella_pragathi 419 Aug 26 03:16 task2_double_quotes.sh
drwxr-xr-x 2 jella_pragathi jella_pragathi 4096 Aug 1 08:01 tests
-rwxr-xr-x 1 jella_pragathi jella_pragathi 16256 Aug 23 16:42 zombie
-rw-r--r-- 1 jella_pragathi jella_pragathi 780 Aug 23 16:41 zombie.c
jella_pragathi@LAPTOP-RQQGT555:~/OSSP$ ls -l output.txt
-rw-r--r-- 1 jella_pragathi jella_pragathi 3782 Sep 17 06:29 output.txt
jella_pragathi@LAPTOP-RQQGT555:~/OSSP$ ./session13_output_redirect
Output Redirection Shell
Example: ls -l > output.txt
output-redirect$ pwd > location.txt
output-redirect$ exit
Output redirection shell terminated.
jella_pragathi@LAPTOP-RQQGT555:~/OSSP$ cat location.txt
/home/jella_pragathi/OSSP
jella_pragathi@LAPTOP-RQQGT555:~/OSSP$ |
```

Skill Session 14

Task 1

To Parse Append Operator, Open Files in Append Mode, Redirect Streams, Preserve Existing Data, Validate Appends, Test Scenarios.



The screenshot shows a terminal window with the nano text editor open. The editor is editing a file named `session14_append_redirect.c`. The code is a C program that implements a simple shell for parsing the append redirection operator (`>>`). It includes standard headers like `<stdio.h>`, `<stdlib.h>`, `<string.h>`, `<unistd.h>`, `<fcntl.h>`, and `<sys/wait.h>`. It defines `MAX_INPUT` as 1024 and `MAX_ARGS` as 50. The `parse_command` function uses `strtok` to split the command into tokens. The `main` function prints a message, an example command, and enters a loop where it prints a prompt and flushes the output.

```
GNU nano 8.7.1 session14_append_redirect.c *
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <fcntl.h>
#include <sys/wait.h>

#define MAX_INPUT 1024
#define MAX_ARGS 50

void parse_command(char *command, char **args)
{
    int i = 0;

    char *token = strtok(command, " \\t\\n");

    while (token != NULL && i < MAX_ARGS - 1)
    {
        args[i++] = token;
        token = strtok(NULL, " \\t\\n");
    }

    args[i] = NULL;
}

int main()
{
    char input[MAX_INPUT];

    printf("Append Redirection Shell\\n");
    printf("Example: echo Hello >> output.txt\\n");

    while (1)
    {
        printf("append-redirect$ ");
        fflush(stdout);
    }
}
```

At the bottom of the terminal, there is a status bar with keyboard shortcuts: `^G Help`, `^X Exit`, `^O Write Out`, `^R Read File`, `^F Where Is`, `^\\ Replace`, `^K Cut`, `^U Paste`, `^T Execute`, and `^J Justify`.

```
jella_pragathi@LAPTOP-RQQQ x + v
GNU nano 8.7.1 session14_append_redirect.c *
    if (fgets(input, sizeof(input), stdin) == NULL)
        break;

    input[strcspn(input, "\n")] = '\0';

    if (strcmp(input, "exit") == 0)
        break;

    char *redirect = strstr(input, ">>");

    if (redirect == NULL)
    {
        printf("Error: Use '>>' for append redirection.\n");
        continue;
    }

    *redirect = '\0';

    char command[MAX_INPUT];
    char filename[MAX_INPUT];

    strcpy(command, input);
    strcpy(filename, redirect + 2);

    char *file = strtok(filename, " \t\n");

    if (file == NULL)
    {
        printf("Error: No output file specified.\n");
        continue;
    }

    char *args[MAX_ARGS];

    parse_command(command, args);

    | if (args[0] == NULL)

^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify
```

```
if (args[0] == NULL)
{
    printf("Error: No command specified.\n");
    continue;
}

pid_t pid = fork();

if (pid < 0)
{
    perror("fork");
    continue;
}

if (pid == 0)
{
    int fd = open(file, O_WRONLY | O_CREAT | O_APPEND, 0644);

    if (fd < 0)
    {
        perror("open");
        exit(EXIT_FAILURE);
    }

    if (dup2(fd, STDOUT_FILENO) < 0)
    {
        perror("dup2");
        close(fd);
        exit(EXIT_FAILURE);
    }

    close(fd);

    execvp(args[0], args);

    perror("execvp");
}
```

^G Help
^X Exit

^O Write Out
^R Read File

^F Where Is
^ Replace

^K Cut
^U Paste

^T Execute
^J Justify

```

        perror("execvp");
        exit(EXIT_FAILURE);
    }
    else
    {
        waitpid(pid, NULL, 0);
    }
}

printf("Append redirection shell terminated.\n");

return 0;
}
|

```

^G Help	^O Write Out	^F Where Is	^K Cut	^T Execute
^X Exit	^R Read File	^N Replace	^U Paste	^J Justify

```

jella_pragathi@LAPTOP-RQQT555:~/OSSP$ cd ~/OSSP
jella_pragathi@LAPTOP-RQQT555:~/OSSP$ nano session14_append_redirect.c
jella_pragathi@LAPTOP-RQQT555:~/OSSP$ gcc -Wall -Wextra -g session14_append_redirect.c -o session14_append_redirect
jella_pragathi@LAPTOP-RQQT555:~/OSSP$ ./session14_append_redirect
Append Redirection Shell
Example: echo Hello >> output.txt
append-redirect$ echo First line >> append.txt
append-redirect$ echo Second line >> append.txt
append-redirect$ echo Third line >> append.txt
append-redirect$ exit
Append redirection shell terminated.
jella_pragathi@LAPTOP-RQQT555:~/OSSP$ cat append.txt
First line
Second line
Third line
jella_pragathi@LAPTOP-RQQT555:~/OSSP$

```

Task 2

To Parse stderr Redirection, Redirect Error Streams, Capture Error Output, Verify File Contents, Handle Failures, Test Error Cases.



```
GNU nano 8.7.1 session14_error_redirect.c *
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <fcntl.h>
#include <sys/wait.h>

#define MAX_INPUT 1024
#define MAX_ARGS 50

void parse_command(char *command, char **args)
{
    int i = 0;

    char *token = strtok(command, " \\t\\n");

    while (token != NULL && i < MAX_ARGS - 1)
    {
        args[i++] = token;
        token = strtok(NULL, " \\t\\n");
    }

    args[i] = NULL;
}

int main()
{
    char input[MAX_INPUT];

    printf("Error Redirection Shell\\n");
    printf("Example: ls missing.txt 2> error.txt\\n");

    while (1)
    {
        printf("error-redirect$ ");
        fflush(stdout);
    }
}
```

Help **Write Out** **Where Is** **Cut** **Execute**
Exit **Read File** **Replace** **Paste** **Justify**

```
if (fgets(input, sizeof(input), stdin) == NULL)
    break;

input[strcspn(input, "\n")] = '\0';

if (strcmp(input, "exit") == 0)
    break;

char *redirect = strstr(input, "2>");

if (redirect == NULL)
{
    printf("Error: Use '2>' for error redirection.\n");
    continue;
}

*redirect = '\0';

char command[MAX_INPUT];
char filename[MAX_INPUT];

strcpy(command, input);
strcpy(filename, redirect + 2);

char *file = strtok(filename, " \t\n");

if (file == NULL)
{
    printf("Error: No error output file specified.\n");
    continue;
}

char *args[MAX_ARGS];

parse_command(command, args);
```

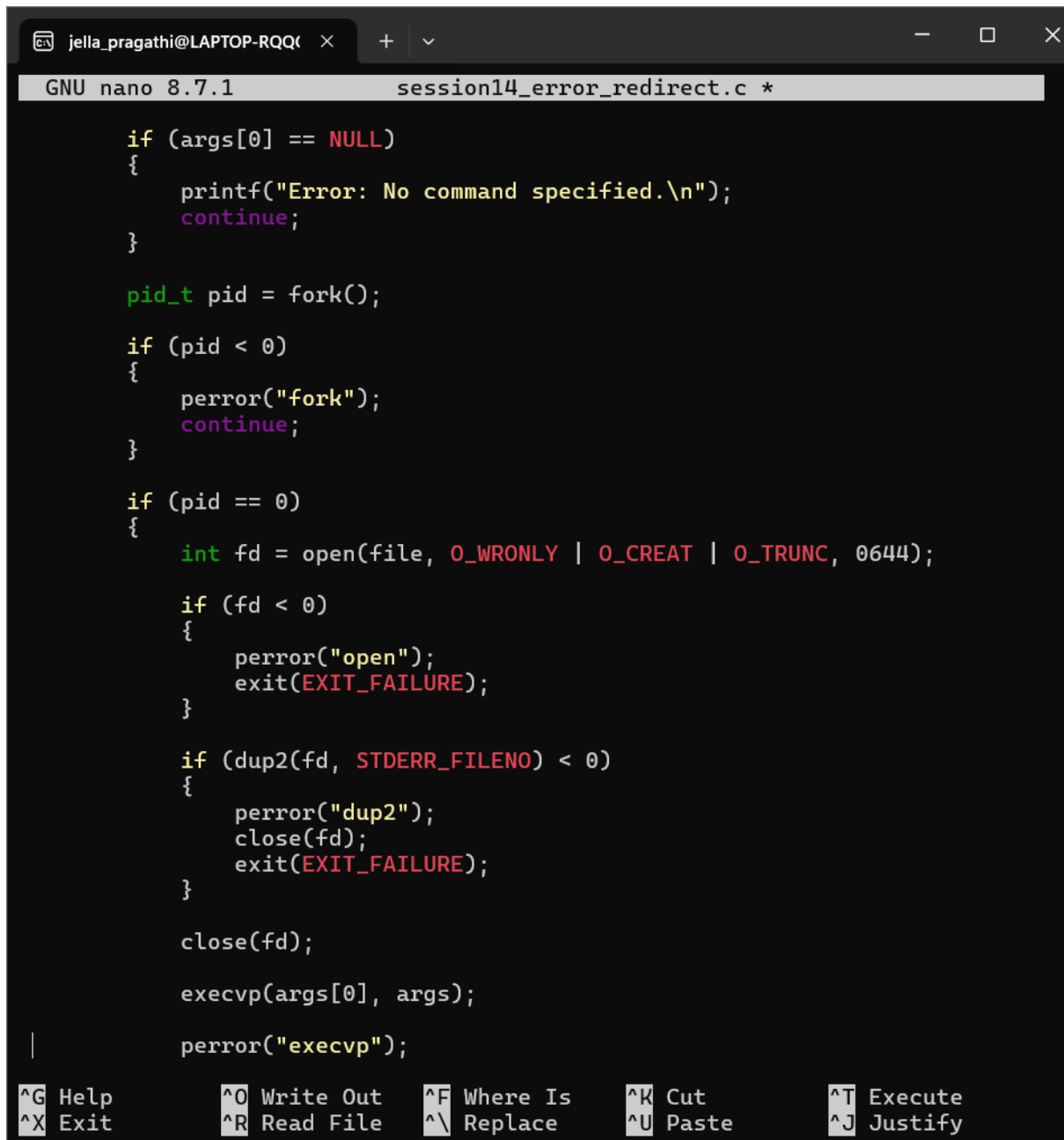
^G Help
^X Exit

^O Write Out
^R Read File

^F Where Is
^ Replace

^K Cut
^U Paste

^T Execute
^J Justify



```
GNU nano 8.7.1 session14_error_redirect.c *

if (args[0] == NULL)
{
    printf("Error: No command specified.\n");
    continue;
}

pid_t pid = fork();

if (pid < 0)
{
    perror("fork");
    continue;
}

if (pid == 0)
{
    int fd = open(file, O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (fd < 0)
    {
        perror("open");
        exit(EXIT_FAILURE);
    }

    if (dup2(fd, STDERR_FILENO) < 0)
    {
        perror("dup2");
        close(fd);
        exit(EXIT_FAILURE);
    }

    close(fd);

    execvp(args[0], args);

    perror("execvp");
}
```

^G Help	^O Write Out	^F Where Is	^K Cut	^T Execute
^X Exit	^R Read File	^_\ Replace	^U Paste	^J Justify

```

        perror("execvp");
        exit(EXIT_FAILURE);
    }
    else
    {
        waitpid(pid, NULL, 0);
    }
}

printf("Error redirection shell terminated.\n");

return 0;
}
|

```

^G Help	^O Write Out	^F Where Is	^K Cut	^T Execute
^X Exit	^R Read File	^_ Replace	^U Paste	^J Justify

```

jella_pragathi@LAPTOP-RQQGT555:~/OSSP$ cd ~/OSSP
nano session14_error_redirect.c
jella_pragathi@LAPTOP-RQQGT555:~/OSSP$ gcc -Wall -Wextra -g session14_error_redirect.c -o session14_error_redirect
jella_pragathi@LAPTOP-RQQGT555:~/OSSP$ ./session14_error_redirect
Error Redirection Shell
Example: ls missing.txt 2> error.txt
error-redirect$ ls missing.txt 2> error.txt
error-redirect$ exit
Error redirection shell terminated.
jella_pragathi@LAPTOP-RQQGT555:~/OSSP$ cat error.txt
ls: cannot access 'missing.txt': No such file or directory
jella_pragathi@LAPTOP-RQQGT555:~/OSSP$ ./session14_error_redirect
Error Redirection Shell
Example: ls missing.txt 2> error.txt
error-redirect$ ls session14_error_redirect.c 2> error2.txt
session14_error_redirect.c
error-redirect$ exit
Error redirection shell terminated.
jella_pragathi@LAPTOP-RQQGT555:~/OSSP$ cat error2.txt
jella_pragathi@LAPTOP-RQQGT555:~/OSSP$ ls -l error2.txt
-rw-r--r-- 1 jella_pragathi jella_pragathi 0 Sep 17 06:38 error2.txt
jella_pragathi@LAPTOP-RQQGT555:~/OSSP$

```